**Conceptual Questions Set 1**

1. A student says, "The genotype is the set of instructions, and the phenotype is the resulting product." Is this analogy fundamentally correct? What crucial element of the genotype-phenotype map does it oversimplify or miss?

2. In a population of genetically identical plants grown in a controlled greenhouse, you observe slight variations in height and leaf number. What does this tell you about the relationship between genotype and phenotype?

3. Why is the concept of a "genotype-phenotype map" more useful for understanding modern genetics than simply memorizing the definitions of "genotype" and "phenotype"?

4. In a standard monohybrid cross (Aa x Aa), the phenotypic ratio in the offspring is 3:1. This simple ratio depends on two critical biological assumptions: One assumption is about the functional relationship between the alleles. The other is about the process of gamete formation and fertilization. What are these assumptions?

5. Mendel's law of segregation is often summarized as "allele pairs separate during gamete formation." Why was the concept of paired alleles that separate a more profound insight than simply observing that traits could be hidden in one generation and reappear in the next?

6. In a dihybrid cross (AaBb x AaBb), the 9:3:3:1 ratio depends on more than just independent assortment. What specific relationship must exist between the two gene pairs (e.g., genes for seed colour and seed shape) for this ratio to hold true?

7. In a population of flowers, red (RR) and white (WW) parents are crossed, and all F1 offspring are pink (RW). A student claims this proves Mendel was wrong about dominance. How would you reframe this discovery to show it actually extends rather than contradicts Mendel's model of inheritance?

8. In human blood types, the $I^A$ and $I^B$ alleles are co-dominant to each other, but both are completely dominant over the i allele. How does this single-gene system demonstrate that "dominance" is not a fixed property of an allele, but a descriptor of its relationship with another specific allele?

9. If co-dominance results in a distinct, blended phenotype (like pink flowers from red and white), how is it different from incomplete dominance?

10. In a pathway for pigment production, Gene A encodes an enzyme to make a colourless precursor, and Gene B encodes an enzyme to convert that precursor to a purple pigment. If a plant is homozygous recessive for Gene A (aa), it will be white, regardless of its genotype at Gene B. Explain why this is a classic example of epistasis, and state which gene is epistatic to the other.

11. How is epistasis fundamentally different from dominance?

12. Epistasis (gene interaction) often modifies classic Mendelian ratios. How does recognizing epistasis change our view of a dihybrid cross from simply "two independent traits" to a more realistic model of how genes function in an organism?

13. In a lab, you find a dihybrid cross yields a 9:7 phenotypic ratio instead of 9:3:3:1. Propose a simple biochemical pathway model that could explain this result.

14. Before Mendel, the "blending theory" of inheritance was popular. How did Mendel's particulate theory (discrete units/genes) solve the two major problems of blending theory: 1) the disappearance of variation, and 2) the reappearance of traits after skipping generations?

15. Mendel knew nothing about DNA, chromosomes, or meiosis. Why is his work considered the foundation of modern genetics despite this? What did he establish that was later physically mapped onto cellular structures and molecules?

16. A critic says, "Mendel's laws are too simplistic; most traits don't follow 3:1 ratios." How would you defend the monumental importance of his work while acknowledging the critic's point about the complexity of many traits?

17. In his ground-breaking 1865 lecture and 1866 paper "Versuche über Pflanzen-Hybriden" (Experiments on Plant Hybrids), Gregor Mendel used the terms "Elementen" (elements/factors) and "Merkmale" (traits/characters) to describe how inherited traits are passed from parents to offspring. Imagine you could explain modern genetics to Mendel. How would you connect his terms "elementen" and "Merkmale" to our modern concepts of gene**,** allele**,** protein**,** and phenotype in a single, coherent sentence?

18. In a monohybrid cross (Aa x Aa), the probability of the first three offspring *all* being homozygous recessive (aa) is $(1/4)^3$. Does the birth of three aa offspring change the probability that the *fourth* child will also be aa? Explain the statistical principle at work.

19. In a dihybrid cross, the 9:3:3:1 ratio is a result of applying the product rule to two independent monohybrid crosses (3:1) x (3:1). What does the *mathematical operation* of multiplication (rather than addition) tell us biologically about how these two traits are being inherited?

20. In classic experiments on coat colour in mice, researchers crossed yellow-bodied mice with grey-bodied mice. All F1 offspring were yellow. When these F1 yellow mice were crossed with each other, they produced: 2 yellow-bodied mice **:** 1 grey-bodied mouse. No true-breeding yellow line could ever be established. Further investigation showed that the yellow allele ($A^y$) is lethal when homozygous ($A^yA^y$); embryos with this genotype do not survive.
    Based on this information:
    i. At the level of coat colour**:** Is the yellow allele ($A^y$) best described as dominant or recessive to the grey allele (A)? Justify your answer with the data.

ii. At the level of viability (survival)**:** Is the yellow allele ($A^y$) best described as dominant or recessive? Justify your answer.

iii. This single allele ($A^y$) affects two very different traits: coat colour and embryonic survival. This is an example of pleiotropy. Pleiotropy is the phenomenon where a single gene influences multiple, seemingly unrelated phenotypic traits**.** What does this single experiment reveal about a critical limitation of describing an allele simply as "dominant" or "recessive" without further context?

iv. Dominance is not an intrinsic property of an allele; it is a *context-dependent descriptor of a relationship between alleles for a specific phenotype*. What do you make of this statement?